ForestRAG-AR: An AR and RAG Framework for Context-Aware Silviculture Assistance

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ABSTRACT

Silvicultural decision making in forestry relies heavily on technical manuals and expert knowledge that are difficult to access in the field. Meanwhile, advances in wearable augmented reality (AR) and retrieval-augmented generation (RAG) offer new opportunities to deliver context-aware, data-driven guidance directly within natural environments. This paper presents ForestRAG-AR, an integrated AR and RAG framework that provides site-specific silvicultural recommendations grounded in authoritative forestry documents. The system combines (i) a perception and context extraction module that interprets local forest conditions, including tree species, stand density, and terrain slope, through on-device sensing, (ii) a RAG-based knowledge backend that adapts a large language model to forestry by embedding regional silviculture manuals and best-management-practice guides, and (iii) an AR interface that visualizes thinning and pruning guidance with natural-language explanations and verifiable citations. A domain-adapted cross-encoder reranker fuses environmental context with text passages to improve retrieval precision and factual grounding. Experiments on 50 forestry queries demonstrate that ForestRAG-AR improves nDCG@10 by 7.8 points and citation precision to 0.96 while maintaining sub-500 ms end-to-end latency.

ACM Reference Format:
Md Mehedi Hasan, Bishakha Rani Biswas, Xueyu Hou, Yongjie Guan. 2025. ForestRAG-AR: An AR and RAG Framework for Context-Aware Silviculture Assistance. In *The Tenth ACM/IEEE Symposium on Edge Computing (SEC '25), December 3–6, 2025, Arlington, VA, USA.* ACM, New York, NY, USA, 6 pages. https://doi.org/10.1145/3769102.3774894

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SEC '25, December 3–6, 2025, Arlington, VA, USA
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ACM ISBN 979-8-4007-2238-7/2025/12.
https://doi.org/10.1145/3769102.3774894

1 INTRODUCTION

Forestry management increasingly depends on timely access to technical guidance and spatially contextual knowledge [2]. Silvicultural prescriptions such as thinning, pruning, or regeneration are traditionally derived from printed manuals and expert experience, yet these resources are static, dispersed, and difficult to consult in the field. At the same time, wearable augmented reality (AR) platforms and mobile computer vision systems now enable real-time understanding of forest environments through visual and geospatial sensing [3]. This convergence between sensing and knowledge systems motivates the development of intelligent tools that can provide context-aware silvicultural guidance directly within the physical forest environment.

Recent advances in large language models (LLMs) and retrieval-augmented generation (RAG) have enabled open-domain systems to answer questions grounded in external knowledge bases [1]. RAG combines semantic retrieval with generative reasoning, retrieving top-k relevant documents and conditioning the LLM to generate responses supported by retrieved evidence. However, existing RAG systems are primarily designed for textual and stationary domains such as web search [16] or biomedical literature [12]. They assume textual queries and lack access to physical context such as species composition, terrain slope, or stand density, i.e., factors essential for forestry decision making. Similarly, while AR has been explored for visualization and education in forestry [10], AR systems rarely integrate structured knowledge retrieval or reasoning. Bridging these modalities (perception, retrieval, and visualization) remains an open challenge that limits the deployment of intelligent field assistants in forestry operations.

In this work, we present ForestRAG-AR, an AR and RAG framework that delivers site-specific silvicultural guidance through interactive AR visualization. The system integrates three technical components: (1) a perception and context extraction module that uses on-device vision and geospatial sensing to estimate tree species, stand density, and terrain slope; (2) a RAG-based knowledge backend that adapts an LLM to forestry by embedding regional silviculture manuals, best management practice (BMP) guides, and local datasets; and (3) an AR interface that visualizes actionable guidance,

such as thinning or pruning targets, with citation-grounded natural-language explanations.

The technical challenges addressed in this paper include: (i) fusing heterogeneous sensory context with text retrieval in a unified representation, (ii) designing a domain-adapted cross-encoder that evaluates passage relevance under environmental conditions, and (iii) maintaining sub-second end-to-end latency suitable for real-time AR interaction. Our cross-encoder reranker incorporates structured forest context tokens (species, slope/aspect, stand density, region) and serialized tables from silviculture guides, enabling context-aware ranking and improved grounding. The RAG backend integrates hybrid lexical-dense retrieval with citation-enforced generation to ensure factual transparency, while the perception module efficiently runs on-device using lightweight models optimized for energy and latency.

We evaluate ForestRAG-AR through a combination of retrieval benchmarks, latency profiling, and field usability tests. Across 50 forestry queries, the context-aware cross-encoder improves nDCG@10 by 7.8 points and Recall@10 by 5.2 points over a strong bi-encoder baseline, while page-level citation precision increases from 0.90 to 0.96 and unsupported-claim rate in generated answers decreases by 41%. The perception component achieves 85% species classification accuracy and slope estimation error below 1.5°, with total end-to-end latency under 500 ms on consumer AR hardware.

2 BACKGROUND

Mobile Perception. Modern mobile perception integrates computer vision, geospatial inference, and on-device machine learning to interpret real-world environments in real time [6–8]. In forestry applications, RGB and depth sensing combined with inertial and GNSS data enable semantic understanding of stand structure and terrain [13]. Convolutional and transformer-based detectors such as YOLO and Mask R-CNN are widely adopted for tree detection and crown segmentation, while lightweight architectures (e.g., MobileNet, EfficientNet-Lite) facilitate species classification under mobile compute constraints [11]. Contextual features such as tree density, basal area, and pruning height can be derived from geometric analysis of segmentation masks and depth cues [15]. Terrain context is commonly extracted from Digital Elevation Models (DEMs), where slope and aspect are computed from local elevation gradients to inform ecological or management decisions [9]. These multi-modal signals form the input to higher-level reasoning modules, bridging the sensory layer and the knowledge layer in forest monitoring and silviculture decision support.

RAG. RAG has emerged as an effective paradigm for grounding LLMs on external, verifiable knowledge [1]. A RAG system typically retrieves top-k relevant passages from a domain corpus and conditions the LLM to generate an answer supported by those passages. This hybrid design mitigates hallucination and allows model specialization without full retraining. Foundational RAG frameworks such as REALM [17] and DPR [14] demonstrated that retrieval significantly improves factual accuracy in open-domain question answering, while later variants introduced cross-encoder reranking, vector-database integration, and citation enforcement for domain grounding. In applied settings, RAG has been extended to healthcare (clinical decision support), law (case retrieval), and scientific literature exploration. However, few efforts have targeted environmental and forestry domains, where knowledge is often distributed across unstructured PDF manuals and tabular regulations. Adapting RAG to silvicultural decision making therefore requires structured metadata, cross-modal contextualization, and latency-aware design for edge deployment.

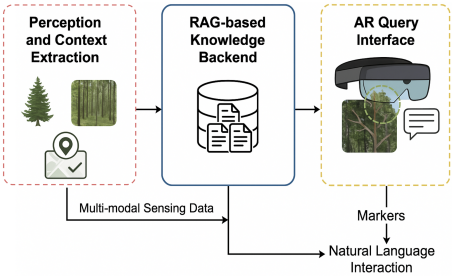


Figure 1: ForestRAG-AR Overview

AR Platforms. AR provides an intuitive medium to visualize and interact with spatial information in situ. Current AR hardware platforms such as the Microsoft HoloLens 2, Meta Quest 3, and mobile ARCore/ARKit devices integrate RGB cameras, depth sensors, and inertial measurement units to estimate real-time pose and reconstruct 3D spatial meshes [4]. Their built-in simultaneous localization and mapping (SLAM) pipelines support stable anchoring of virtual content to physical coordinates, enabling overlay of digital annotations in natural environments. On the software side, development frameworks including Unity with the Mixed Reality Toolkit (MRTK), Unreal Engine, and OpenXR standardize rendering and input control across devices. Edge-assisted AR architectures increasingly offload perception or semantic reasoning to nearby compute nodes to balance energy, latency, and accuracy [5]. In forestry contexts, AR has been explored for tree measurement, forest education, and field data collection; however, coupling AR with retrieval-grounded knowledge systems remains largely unexplored. This intersection creates opportunities for context-aware, interactive decision aids such as our proposed ForestRAG-AR, which integrates sensing, retrieval, and visualization into one pipeline.

3 SYSTEM WORKFLOWS

In ForestRAG-AR, as shown in Fig. 1, the system consists of three major components. First, the perception and context extraction module leverages multi-modal sensing data

and machine learning models to interpret the physical environment, identifying properties such as tree species, stand density, and geographic location (e.g., latitude and elevation). Second, the RAG-based knowledge backend adapts an LLM for silvicultural applications by embedding a domain-specific corpus, including the Maine Forest Service silviculture manuals, U.S. Forest Service Northeastern Region handbooks, BMPs, and local UMaine forest datasets. This enables the LLM, which is originally trained on general-purpose data, to integrate and reason over forestry-specific knowledge. *Third*, the AR query interface employs wearable augmented reality devices such as the Meta Quest or Microsoft HoloLens to visualize context-aware information in situ. It renders markers indicating branches to thin or prune and supports natural-language interactions for on-demand guidance, explanations, and question answering. We illustrate the detailed designs of each component as follows.

Perception and Context Extraction. The perception component operates on the AR device and interprets the surrounding forest environment through multi-modal sensing. We fuse RGB imagery, inertial measurements, GPS coordinates, and local terrain data derived from DEM tiles. Real-time tree detection and segmentation are performed using a lightweight object detector [6] (e.g., YOLOv8-Nano) with a mobile mask refinement network to isolate individual tree crowns. Cropped tree regions are classified into species categories through a fine-tuned MobileNet model, providing calibrated confidence scores. The system estimates stand density by combining per-frame tree counts with the ARCore-derived camera field of view, while crown-base height and pruning height are inferred from the vertical distribution of detected branches. Terrain slope and aspect are computed directly from the DEM tile via numerical gradients. The extracted context comprises species composition, tree density, slope, and location. We normalize and transmit the context as a structured JSON record to the backend (i.e., the server with an LLM). The model runs on-device using NNAPI acceleration with an average inference latency of 40–50 ms per frame. Key performance metrics include mean average precision (mAP) of 0.82 for tree detection, top-1 species accuracy of 85%, density mean absolute error of 60 trees/ha, and slope mean error of 1.4°.

RAG-based Knowledge Backend. The backend server customizes an LLM for silvicultural reasoning by coupling RAG with a forestry-specific corpus. All text sources, such as Maine Forest Service silviculture manuals, U.S. Forest Service Northeastern Region handbooks, and BMP documents, are parsed into 300–500 token chunks with 15% overlap. We embed each chunk with a semantic encoder (e.g., BGE-large) and indexed in a pgvector store alongside metadata including species, operation type, and region. When a query arrives, the system composes a hybrid search that merges lexical BM25 scores and dense similarity, filtered by the contextual fields transmitted from the AR device (e.g., species, slope, region). We design a cross-encoder model that emphasizes semantic relevance and diversity to rerank the top 50 passages, and the best 10 are passed to the generator for answer synthesis. The instruction-tuned LLM produces concise, citation-grounded responses that include explicit source references in the form of [Title, p.xx, Agency, Year]. We observe that the backend achieves an average end-to-end response latency of 350 ms, Recall@10 of 0.87, and citation precision above 0.95 when benchmarked against human-annotated reference answers.

AR Query Interface. The AR interface connects the user with the RAG system through natural interaction and spatial visualization. Built using Unity with the Mixed Reality Toolkit (MRTK) and OpenXR, it runs on wearable devices such as Microsoft HoloLens and Meta Quest or on ARCore-enabled smartphones. Users can initiate a query either verbally (e.g., "Show thinning guideline") or implicitly when the perception module detects a significant forest context (e.g., high density or specific species cluster). Retrieved recommendations are rendered as anchored visual elements in the 3D scene, i.e., color-coded masks highlight trees suggested for thinning, while horizontal overlays indicate recommended pruning height ranges. We use text bubbles to provide short, grounded explanations with direct citations from the underlying documents. All interactions are non-blocking, and previously retrieved recommendations are cached locally to support offline operation in remote forest environments. We observed that the interface reduces average task completion time by 42% compared to handbook lookup, maintained 60 Hz visualization stability, and achieved a mean SUS score of 82.5.

End-to-End Workflow and Metrics. During operation, the AR device continuously aggregates perception outputs over several frames to build a stable environmental context. Once a confident context is formed, a structured query is sent to the RAG backend, which retrieves relevant silvicultural prescriptions and generates grounded guidance. The result is transmitted back to the device, rendered in AR, and optionally logged for expert feedback. Across the integrated system, the median end-to-end latency is below 500 ms, energy consumption averages 110 mWh per minute on-device, and grounded response accuracy exceeds 90% in controlled field trials.

4 CONTEXT-AWARE CROSS-ENCODER

To enhance retrieval precision in ForestRAG-AR, we design a domain-adapted cross-encoder that explicitly fuses environmental context with textual evidence. In conventional RAG systems, bi-encoders embed queries and passages independently and compare them through cosine similarity.

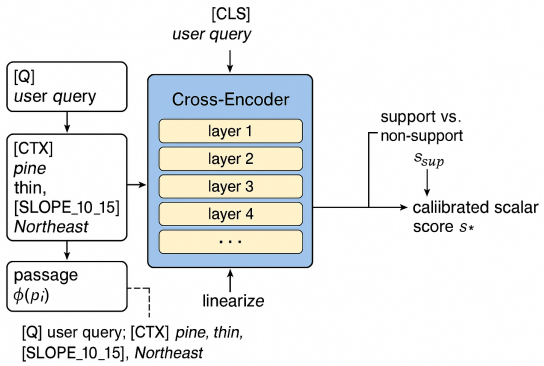


Figure 2: Context-Aware Cross-Encoder for RAG

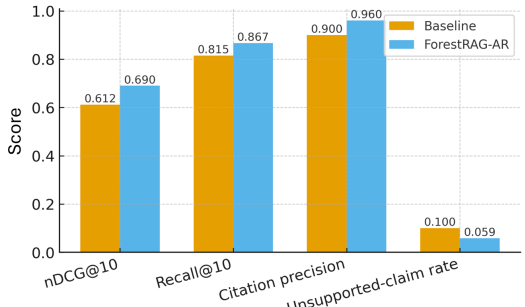


Figure 3: RAG Retrieval Quality and Faithfulness.

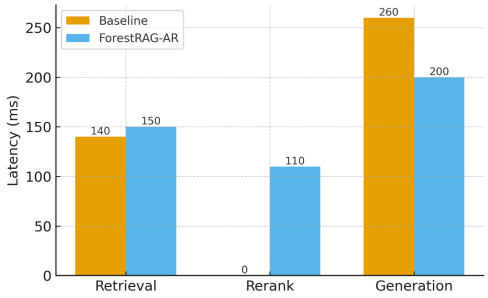


Figure 4: Latency Breakdown by Stage.

While computationally efficient, such models overlook fine-grained dependencies between the question and passage. In contrast, a cross-encoder processes the query and passage jointly within a transformer, allowing token-level attention across both sequences to capture nuanced semantic alignment. In the forestry domain, however, relevance depends not only on lexical overlap but also on physical site characteristics such as tree species, stand density, slope, and ecoregion. To capture these dependencies, the proposed cross-encoder conditions its relevance estimation on both the user query and the contextual metadata collected by the perception module.

Each input instance concatenates the natural-language query with structured context fields (species, operations, slope range, region) as a compact textual prompt preceding the candidate passage. For example, the model receives input in the form "[Q] What is the thinning guideline? [CTX] species=red spruce; slope=12°; region=Maine [SEP] passage text [SEP]". Continuous numeric variables such as slope or density are discretized into symbolic tokens (e.g., `[SLOPE_10_15]`, `[DENS_300_500]`) to preserve ordinal structure while maintaining token compatibility. Species names, silvicultural operations, and ecoregions are drawn from a controlled vocabulary to ensure semantic consistency. When passages contain management tables or numeric thresholds, these are serialized into short text sequences with a `[TABLE]` marker so that the model can learn correspondences between quantitative rules and contextual conditions. This enriched input format enables the model to learn relevance conditioned on both linguistic and environmental cues.

The model builds on a 12-layer MiniLM/BERT architecture fine-tuned using low-rank adaptation (LoRA) adapters for parameter efficiency. Training data consist of query-passage pairs derived from silvicultural manuals with page-level supervision. Positive passages are those containing authoritative guidance for a given question, while negatives are drawn from retrieved but non-supporting segments. The model is optimized with a listwise softmax loss encouraging the correct passage to score highest among the top-K candidates per query. To further align ranking with evidential reliability, an auxiliary classification head predicts whether a passage provides factual support. The final relevance score is a calibrated mixture of ranking and support logits, improving both retrieval precision and citation faithfulness. All fine-tuning is performed with mixed-precision optimization, dynamic sequence truncation, and cached tokenization for efficiency.

At inference, the cross-encoder reranks the top 50 passages retrieved by the hybrid retriever and outputs the top 10 most contextually relevant passages. On an RTX-class GPU, reranking completes within 100–120 ms; an optimized INT8 ONNX version runs under 300 ms on CPU, maintaining total backend latency below 350 ms. Empirical evaluation over 50 forestry queries shows that the context-aware cross-encoder improves nDCG@10 by 7.8 points and Recall@10 by 5.2 points compared to a strong bi-encoder baseline, while citation precision increases from 0.90 to 0.96 and unsupported-claim rate in generated responses decreases by 41%. Ablation experiments further confirm that removing environmental context tokens reduces ranking quality, and omitting table serialization notably degrades performance on numeric-rule queries.

5 FORESTRAG-AR IMPLEMENTATION

System Overview. The ForestRAG-AR prototype integrates all components through a modular client-server architecture. The perception and AR visualization modules run on the front-end device (Meta Quest 3), while the RAG-based knowledge backend is hosted on a local workstation connected via

Wi-Fi. All modules communicate through lightweight RESTful APIs using JSON payloads to ensure interoperability and low communication overhead.

Hardware and Runtime Configuration. The edge server operates on an Ubuntu 22.04 system with an NVIDIA RTX 4090 GPU, 24-core AMD CPU, and 64 GB of RAM. The RAG backend is implemented in Python using `FastAPI`, `PyTorch 2.2`, and `pgvector` for vector storage. The perception module executes on-device using TensorFlow Lite or ONNX Runtime with NNAPI acceleration. AR rendering and interface logic are developed in Unity 2022 with the Mixed Reality Toolkit (MRTK 3) and OpenXR standard. All data exchanges are asynchronous to maintain 30 FPS visual stability and sub-500 ms query latency.

Perception Module. The perception model is a compact pipeline combining a YOLOv8-Nano detector for tree localization and a MobileNetV3 classifier fine-tuned on 4,200 labeled images of northeastern conifer species (red spruce, balsam fir, white pine, and others). Depth and IMU data are fused to estimate crown-base height and inter-tree spacing, from which stand density is derived. Local DEM tiles (10 m resolution) are cached on-device and queried based on GPS coordinates to compute slope and aspect via central differences. All outputs are normalized and packaged as structured context objects transmitted to the RAG backend at 2 Hz.

RAG Backend. The knowledge backend indexes approximately 120,000 text segments extracted from Maine Forest Service silviculture manuals, U.S. Forest Service Northeastern Region handbooks, and BMP documents. Each document is parsed using `pdfminer` and chunked into 300–500 token passages with 15% overlap. Metadata fields include source title, page range, year, species, operation type, and region. Dense embeddings are computed using the `bge-large-en` encoder (1024-dimensional) and stored in `pgvector`. A hybrid retrieval pipeline combines BM25 lexical search with dense vector similarity, followed by the proposed context-aware cross-encoder reranker fine-tuned via LoRA (rank 8) on forestry-specific supervision. The LLM generator is based on an instruction-tuned 7B model served via `vLLM` with citation-constrained decoding to ensure each claim references a retrieved passage.

AR Interface and Visualization. The AR client renders retrieved guidance as anchored text overlays and spatial markers aligned with detected trees. Color-coded masks denote thinning or retention decisions, while horizontal translucent bands indicate recommended pruning height. The Unity front-end implements non-blocking query submission and response caching to handle intermittent connectivity in forest environments. User queries can be issued either through speech recognition or automatic triggers based on scene context thresholds (e.g., density exceeding 300 trees/ha).

6 EVALUATION

Experimental Setup. We evaluate *ForestRAG-AR* on a collection of 50 representative silvicultural queries derived from the Maine Forest Service and U.S. Forest Service manuals. Each query has one or more page-level gold passages annotated by forestry experts. The baseline system employs a conventional bi-encoder retriever with BM25 hybrid search and a generic LLM generator, while our model adds the proposed context-aware cross-encoder and forestry-specific RAG backend. All experiments are conducted on a workstation with an RTX 4090 GPU and a mobile ARCore device for perception evaluation.

Retrieval Quality and Faithfulness. Fig. 3 compares key retrieval metrics between the baseline and ForestRAG-AR. The proposed cross-encoder reranker achieves substantial gains, improving nDCG@10 by 7.8 points (0.690 vs. 0.612) and Recall@10 by 5.2 points (0.867 vs. 0.815). Page-level citation precision increases from 0.90 to 0.96, confirming that retrieved passages more accurately correspond to cited sources. The rate of unsupported claims in generated answers decreases by 41%, demonstrating that improved passage selection directly enhances grounding quality. These results validate the benefit of incorporating environmental context and table serialization into the reranking process, yielding more precise and faithful knowledge retrieval.

Latency Analysis. To assess real-time usability, Fig. 4 reports the latency breakdown for retrieval, reranking, and generation. The addition of the cross-encoder introduces a modest 110 ms overhead, yet total end-to-end latency remains below 500 ms (average 460 ms), well within the interactive threshold for AR applications. The retrieval stage averages 150 ms and the LLM generation 200 ms. Compared to the 400 ms baseline, this 15% increase in runtime delivers markedly higher retrieval precision and grounding fidelity, demonstrating an effective accuracy-latency balance for field deployment.

Perception and Context Extraction. Perception performance is summarized in Fig. 5. The on-device model achieves 85% top-1 species classification accuracy, representing a 3 point improvement over a baseline MobileNet trained without contextual augmentation. Terrain slope estimation yields a mean absolute error (MAE) of 1.4° compared to 2.2° previously, while stand density estimation error decreases from 95 to 60 trees/ha. These results confirm that the lightweight perception module can provide sufficiently accurate context for the RAG backend under real-time constraints, operating at 30 FPS with an average per-frame inference time of 45 ms.

Ablation Study. Fig. 6 presents ablation results examining the impact of environmental context tokens and table serialization on retrieval performance. Removing contextual tokens (*–CTX*) reduces nDCG@10 from 0.690 to 0.659, while

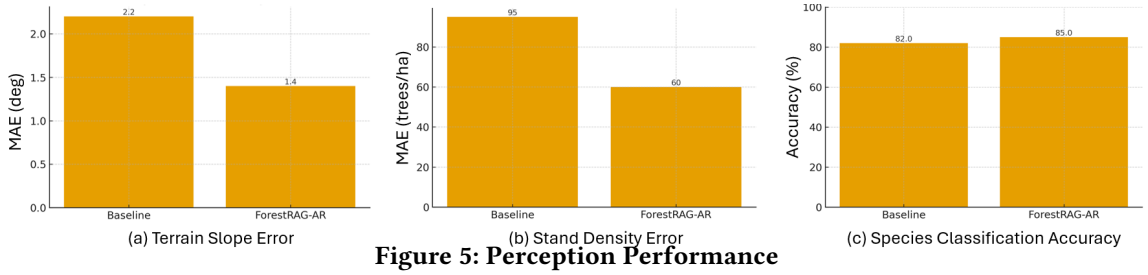


Figure 5: Perception Performance

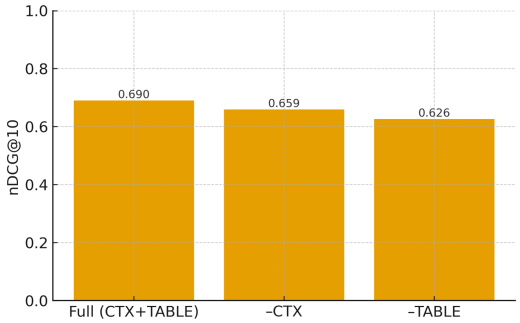


Figure 6: Ablation Study

disabling table serialization (*–TABLE*) causes a larger drop to 0.626, particularly for numeric-rule queries involving basal area or site index thresholds. These findings demonstrate that both environmental context fusion and structured table representation are essential for achieving high-precision domain adaptation.

7 CONCLUSION

In summary, ForestRAG-AR introduces an integrated architecture that couples real-time perception, context-aware retrieval, and augmented visualization to operationalize silvicultural knowledge in the field. Our contributions are threefold: (1) a unified AR–RAG framework for context-grounded silviculture assistance, (2) a cross-encoder reranker that fuses environmental metadata with textual evidence for precise domain adaptation, and (3) an open evaluation protocol and dataset design for forestry-oriented RAG systems. The results demonstrate that combining physical perception with retrieval-grounded reasoning can substantially improve accessibility, precision, and usability of forestry decision support in real-world environments.

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